

Investigation - Conduction

Write up the following investigation using the standard Keynote or Pages.

INQUIRY: INVESTIGATION 10.2

Heat conduction in solids

KEY INQUIRY SKILLS:

- processing and analysing data and information
- evaluating

Equipment:

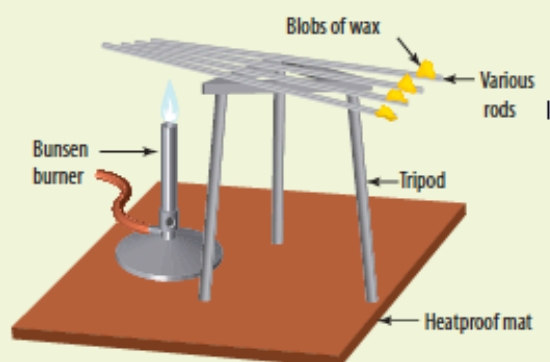
set of three or four metal rods, different metals but identical in size; or heat conduction apparatus

wax candle Bunsen burner and heatproof mat

matches tripod

ruler stopwatch

- Set up the tripod and rods or heat conduction apparatus as shown below.
- Light the candle and melt a blob of wax onto one end of each rod. Ensure that each wax blob is the same distance from the end that will be heated by the Bunsen burner flame.



- Use the blue flame of the Bunsen burner to heat the end of each rod. Start the stopwatch at the instant that heating begins. Record the time taken for each blob to produce its first droplet of wax.
- Present your data in a table and draw a bar or column graph in which to display them.

DISCUSS AND EXPLAIN

- 1 According to your data, which of the metals is the best conductor of heat?
- 2 According to your data, which of the metals is the poorest conductor of heat?
- 3 Compare your data with that of others in your class. Comment on the consistency of the conclusions within your class. If there was inconsistency, suggest one or more reasons for it.

